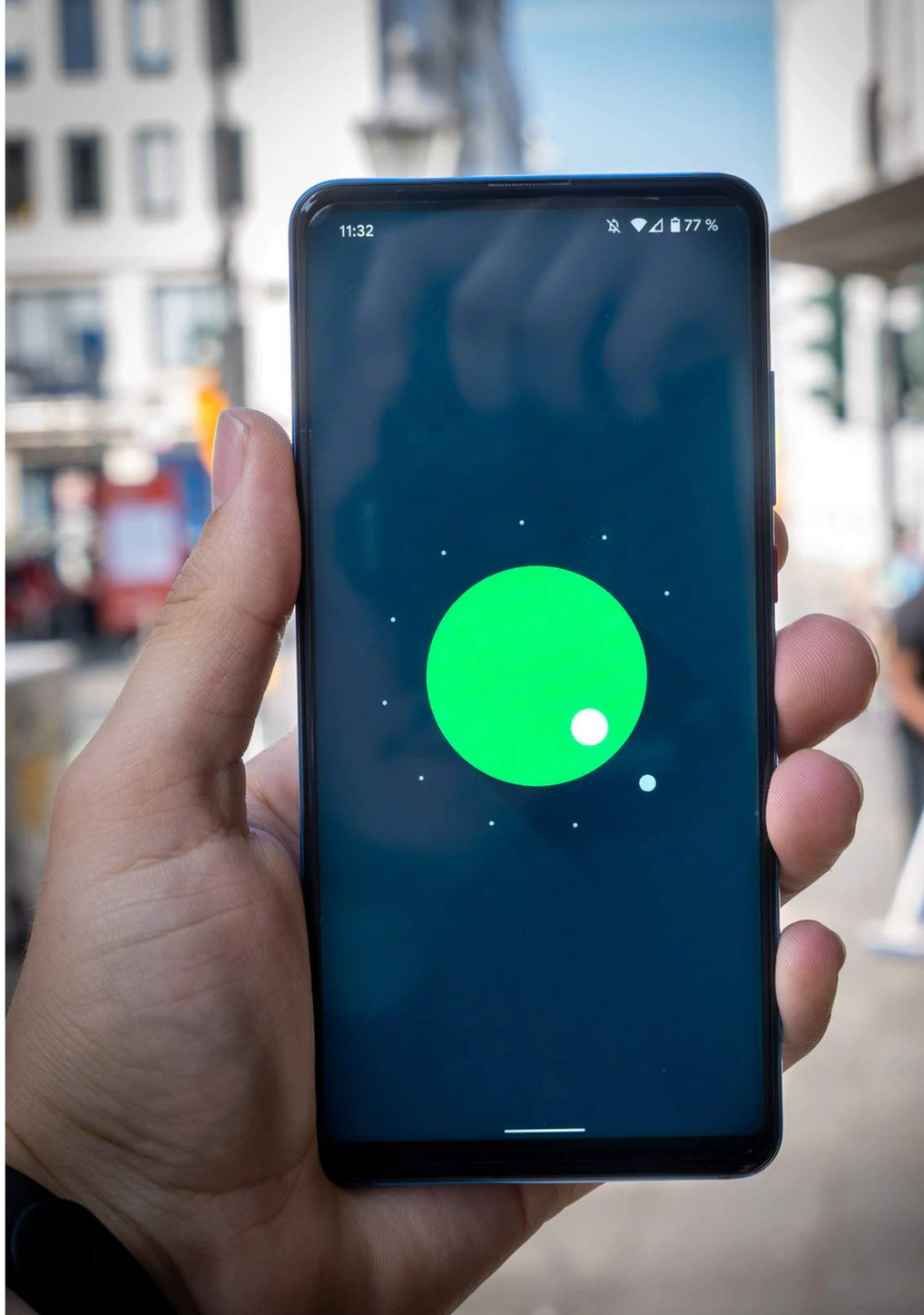
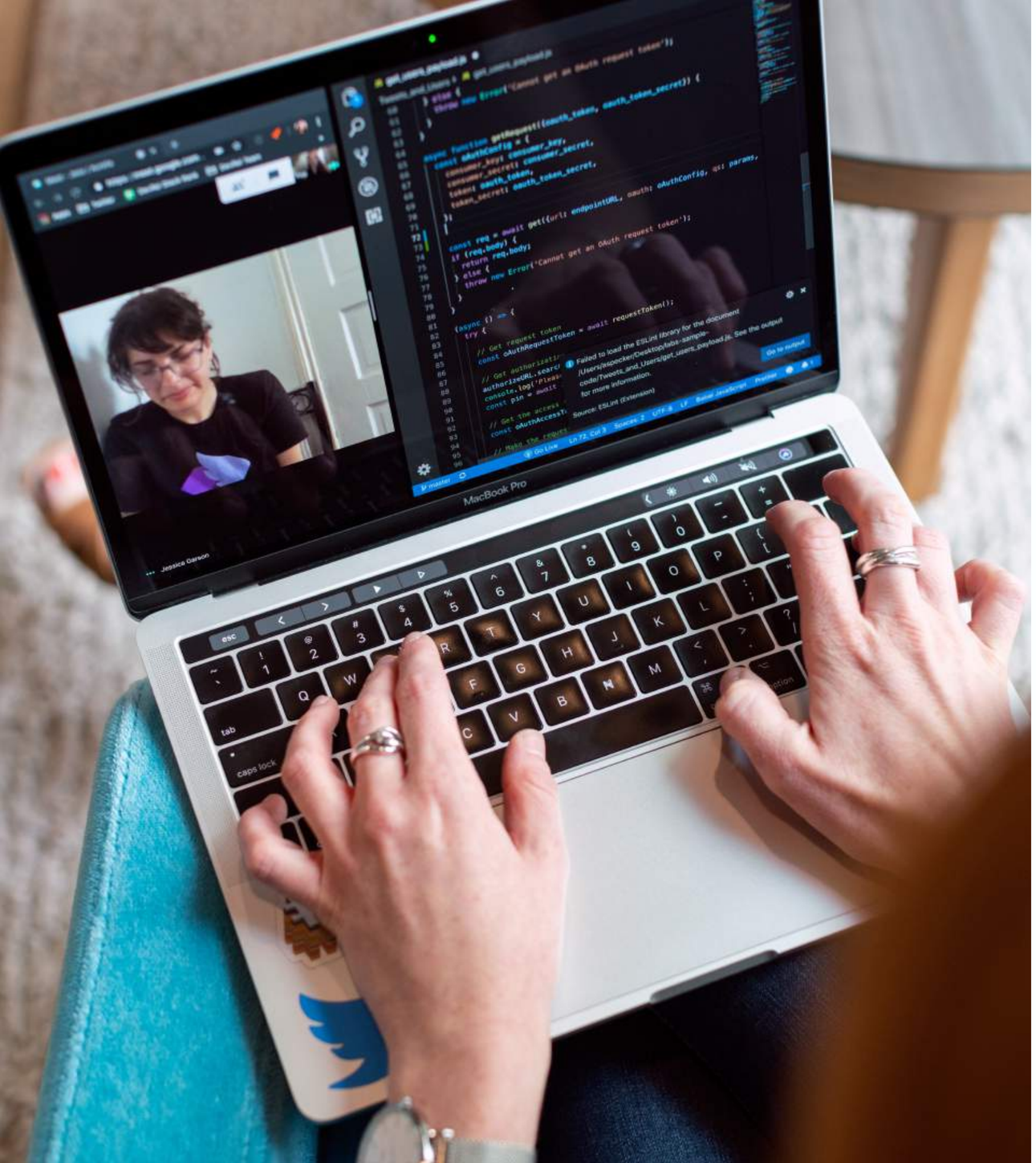
Android Forensics

: Lock screen in old Android device

2418 배상혁 / 21.08.15



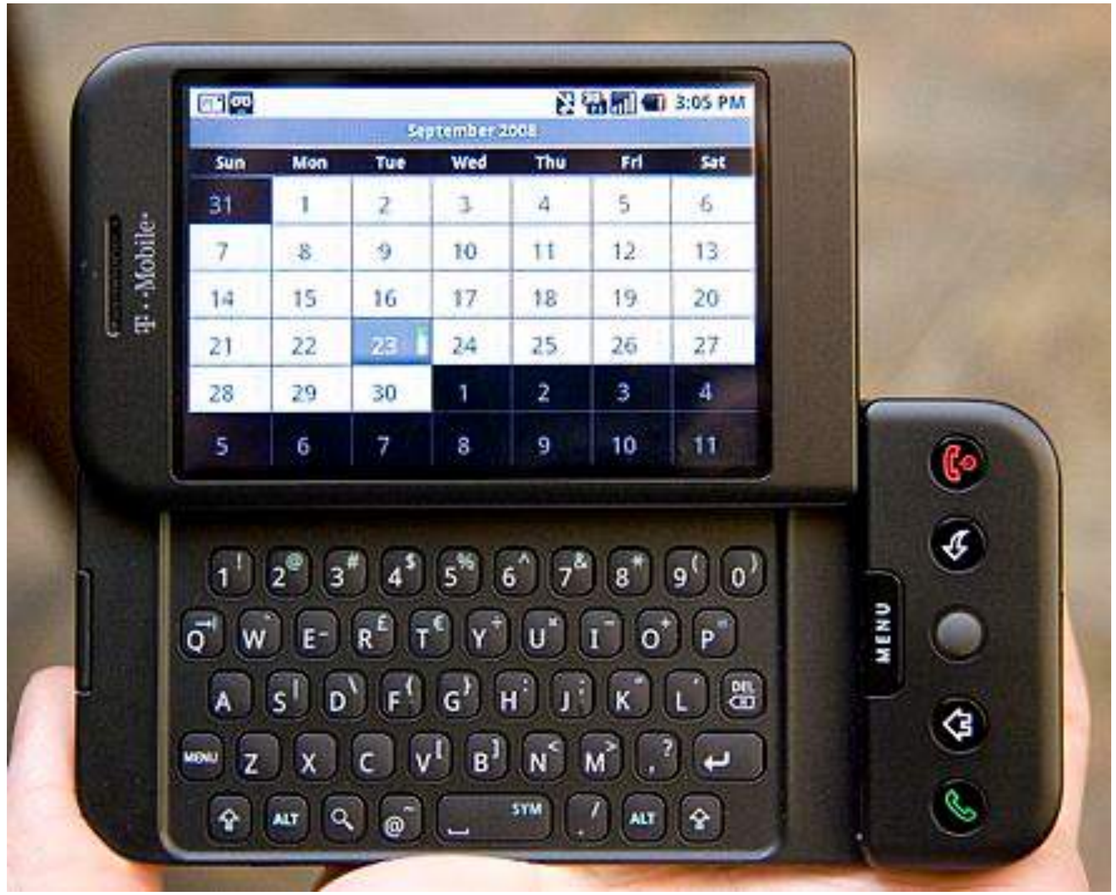


Contents

1. History of Android
2. Installation
3. Several ways to root Android device
4. How to get rid of lock screen
5. Crack pattern and pin
6. Why old Android?
7. Problem solving

History of Android

- Beta (07.11.04)
- Android 1.0 G1 (08.09.23)
- Android 1.1



History of Android



Android
Cupcake



Donut
Android 1.6

1.5 Cupcake

1.6 Donut

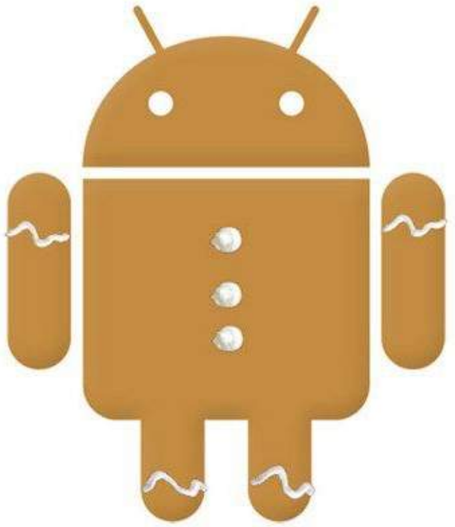
History of Android



2.0 ~ 2.1 Eclair

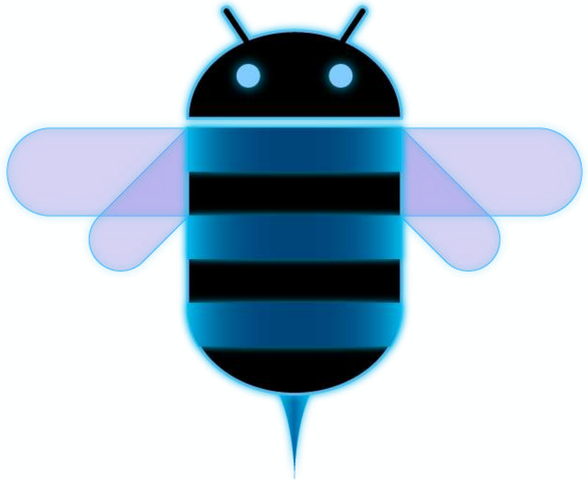


2.2 Froyo



2.3 Gingerbread

History of Android



3.0 ~ 3.2 Honeycomb



4.0 Ice cream Sandwich



4.1 ~ 4.3 Jelly Bean



4.4 Kitkat

History of Android



5.0 Lollipop



6.0 Marshmallow



7.0 ~ 7.1 Nougat

History of Android



8.0 ~ 8.1 Oreo



9.0 Pie

History of Android



Android 10



Android 11



Android 12

Installation





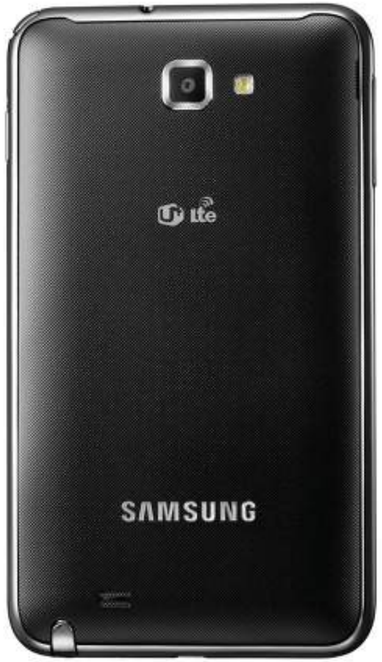




Installation

Old Android Device







Devices with Android version less than 4.3

Installation

ADB

Android Debug Bridge

Commonly abbreviated as ADB is a tool to run commands on a connected Android device

Download

https://developer.android.com/studio/releases/platform-tools



Installation

Odin

- Odin is a utility software developed and used by Samsung internally. It can be used to flash a custom recovery firmware image (as opposed to the stock recovery firmware image) to a Samsung Android device. Odin is also used for unbricking certain Android devices.

- There is a program called jodin, odin for mac, but not compatible with apple silicon.

- http://bit.do/untrust-android-odin

Installation

DB Browser for SQLite

- A program to manage the sqlite3 database conveniently in the GUI environment
- https://sqlitebrowser.org/

Several ways to root Android device

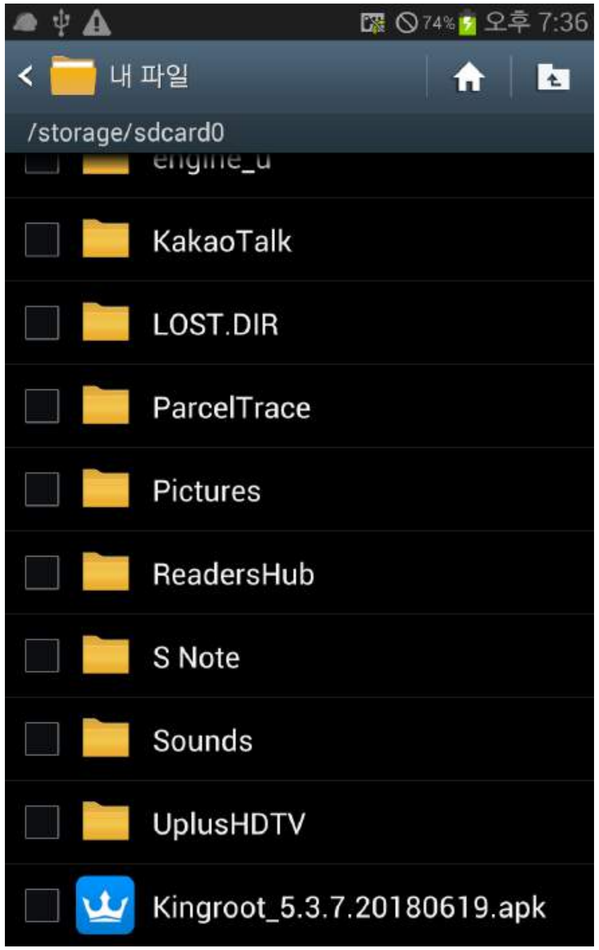
1. KingRoot

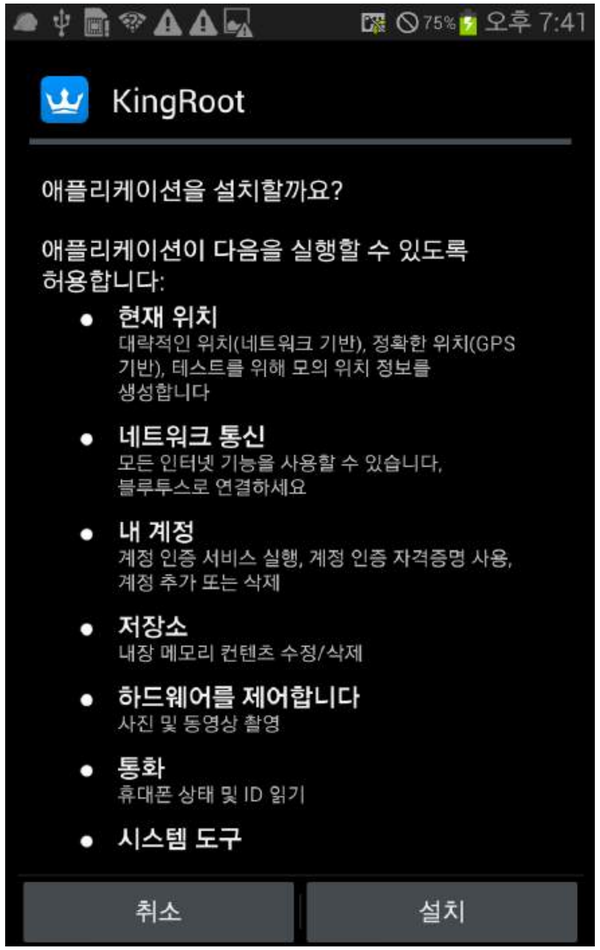
- An app that allows you to root easily with just a few touches
- Download via PC and push to the Android device using adb
- http://bit.do/untrust-android-kingroot

```
~/Downloads/platform-tools                                    at 14:53:03
❯ ./adb push /Users/devleo/Downloads/Kingroot_5.3.7.20180619.apk /sdcard
/Users/devleo/Downloads/Kingroot_5.3.7.20180619.apk: 1 file pushed, 0 skipped. 4.0 MB/s (13232536 bytes in 3.153s)
```

Several ways to root Android device

1. KingRoot



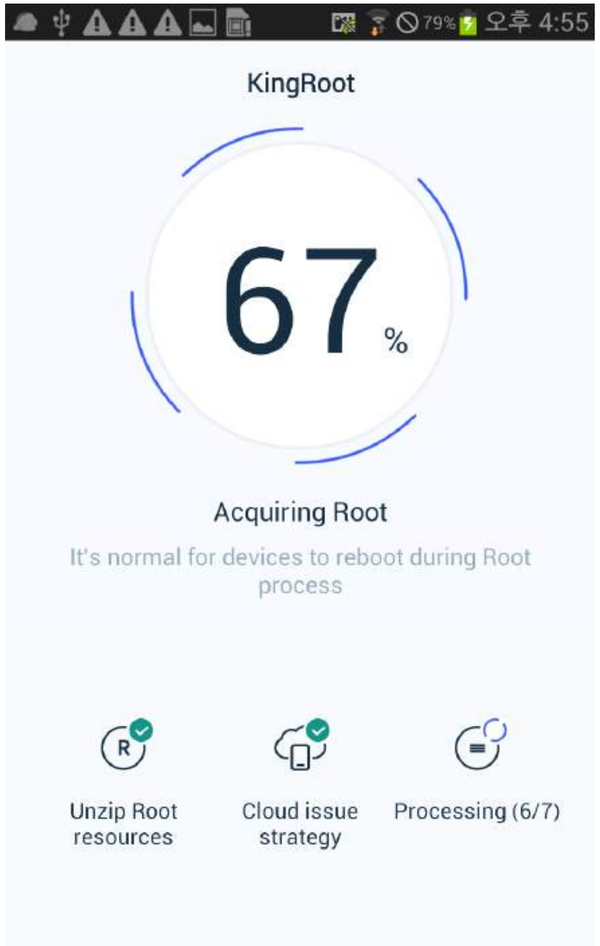




Several ways to root Android device

1. KingRoot





Several ways to root Android device

2. Odin

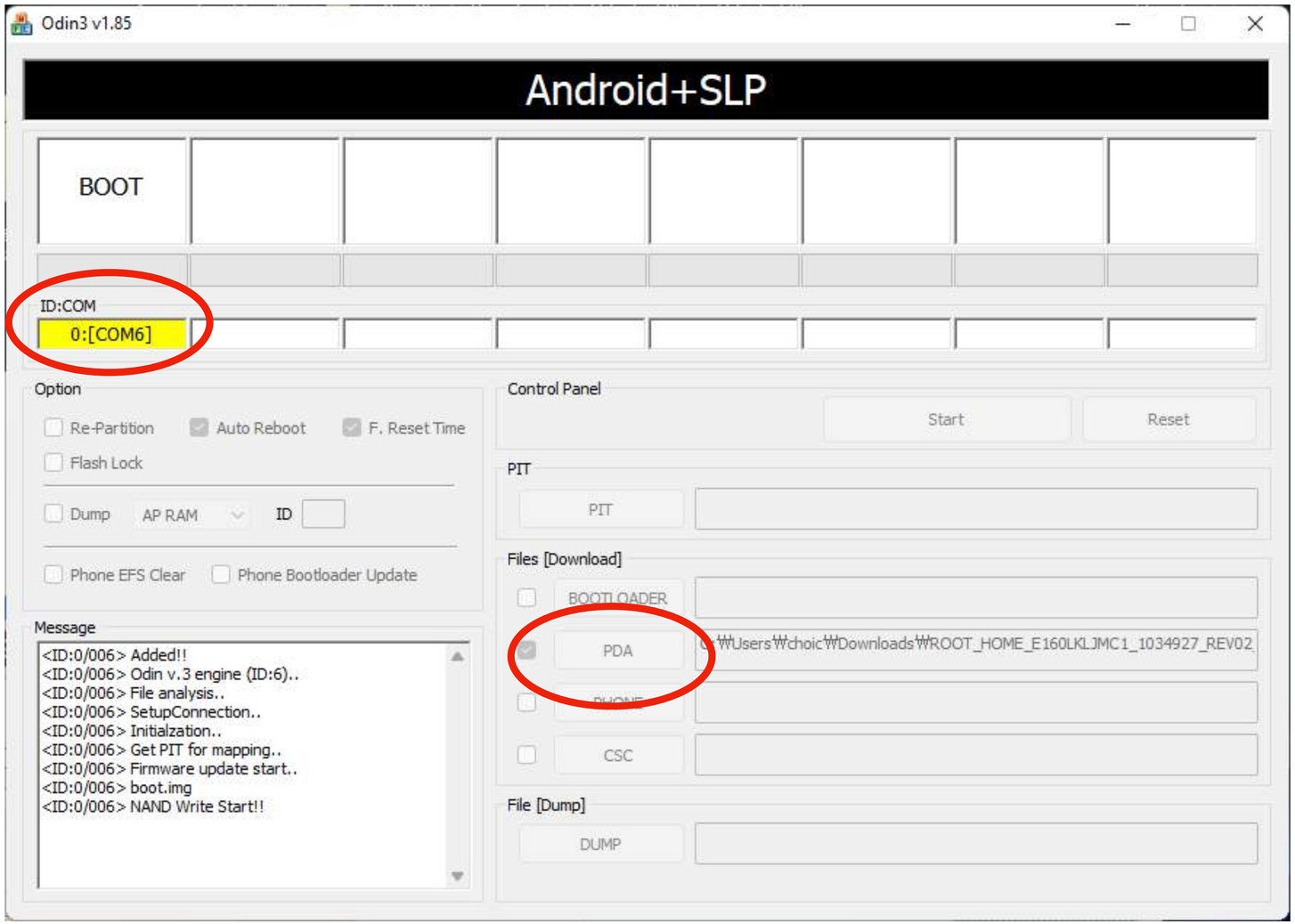
1. Download rooted firmware
http://www.matcl.com/torrent/2656028

2. Boot into recovery

3. Click PDA and insert firmware

4. Check device connection
(ID:COM)

5. Flash start



Several ways to root Android device

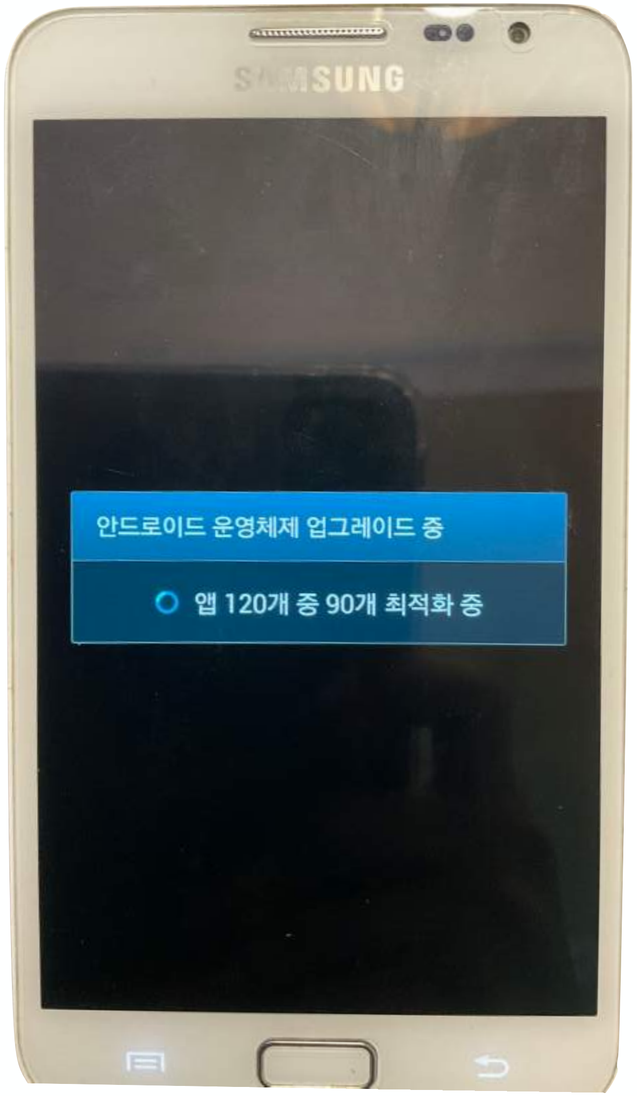
2. Odin





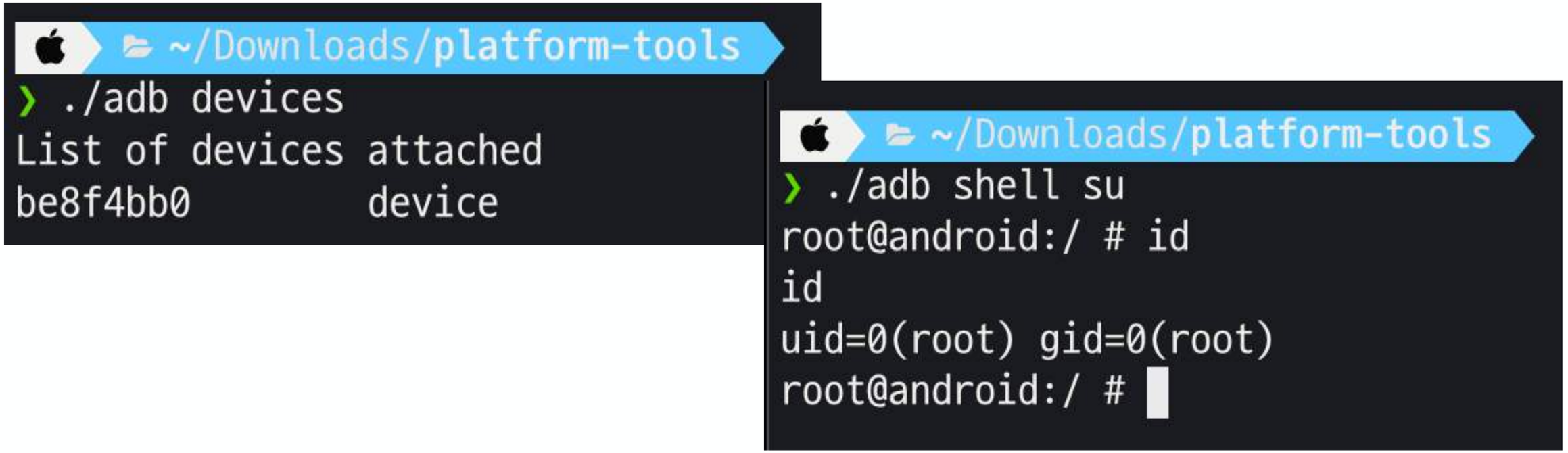
Several ways to root Android device

2. Odin





How to get rid of lock screen



root success !

How to get rid of lock screen

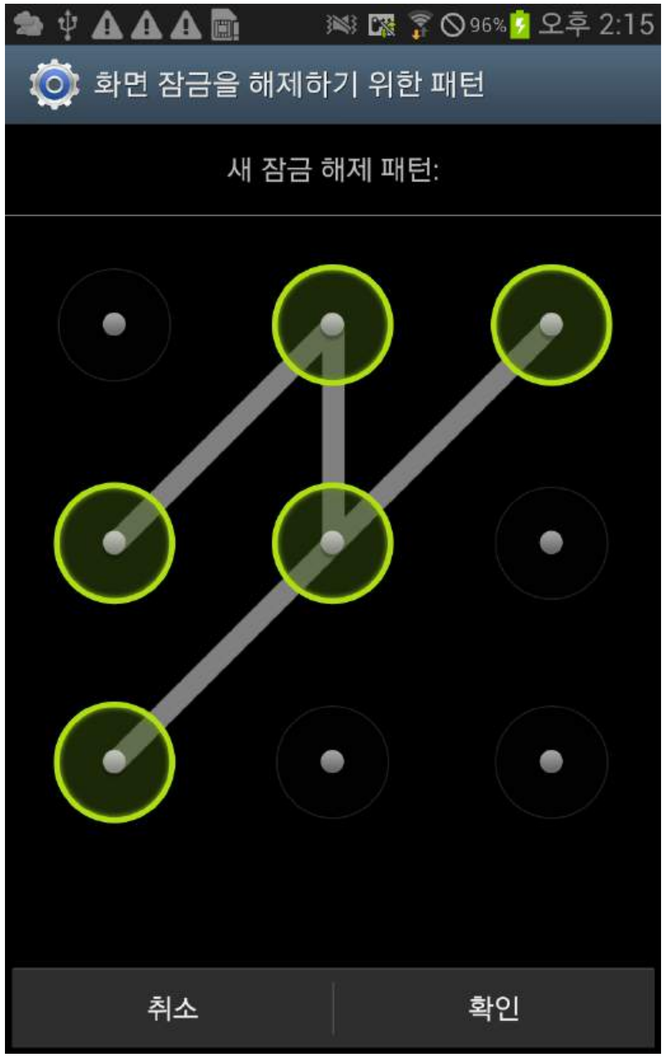
```
root@android:/data/system # rm gesture.key
rm gesture.key
root@android:/data/system #
```

Delete /data/system/gesture.key



Crack pattern and pin

Pattern



Crack pattern and pin

Pattern

```
~/Downloads/platform-tools
❯ ./adb pull /data/system/gesture.key .
adb: error: failed to copy '/data/system/gesture.key' to './gesture.key': remote Permission denied

~/Downloads/platform-tools
❯ ./adb shell "su -c cat /data/system/gesture.key" > gesture.key

~/Downloads/platform-tools
❯ cat gesture.key
[illegible]

~/Downloads/platform-tools
❯
```

Root privileges cannot be used for “adb pull”, so use the following command.

Crack pattern and pin

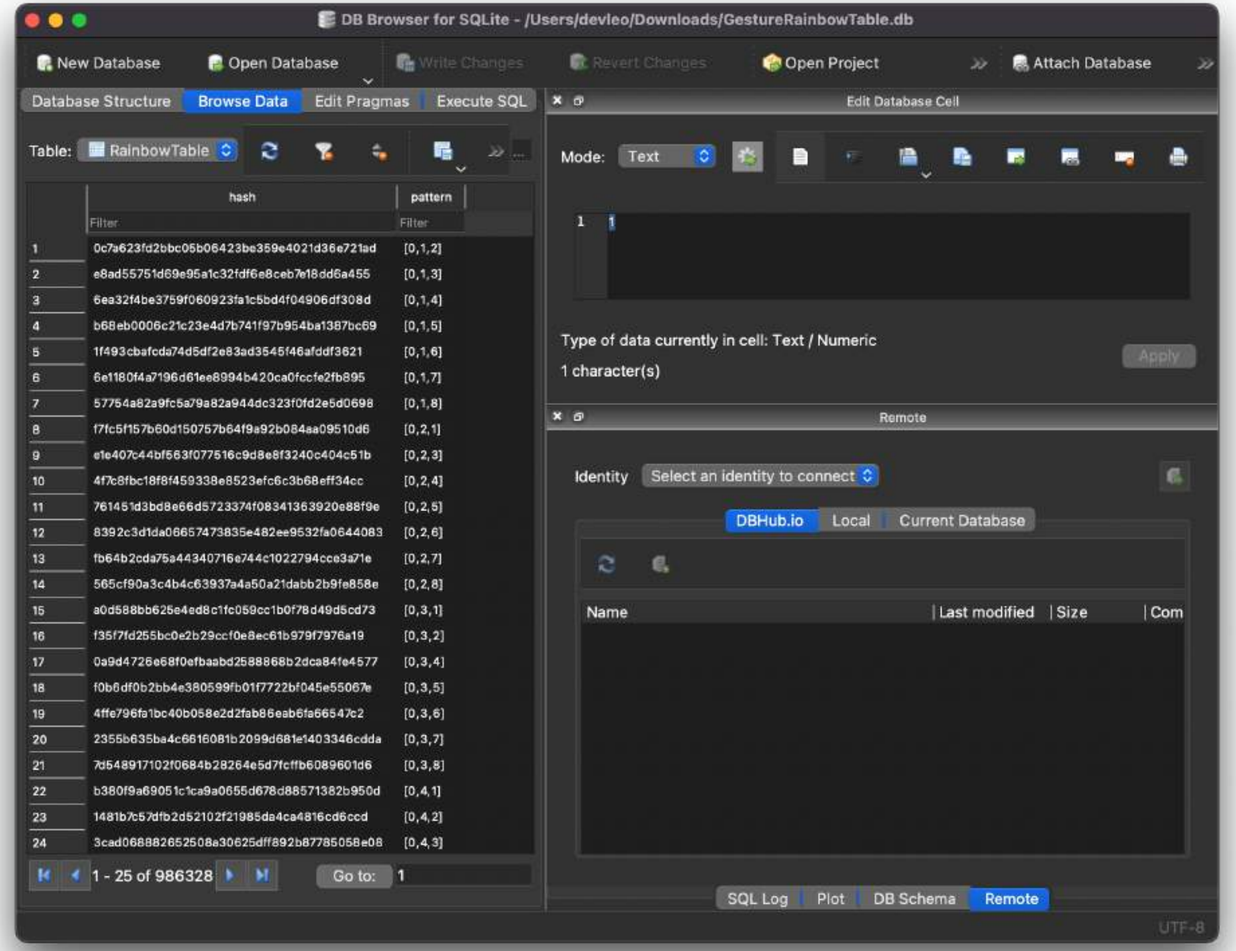
Pattern



SHA-1 encrypted value

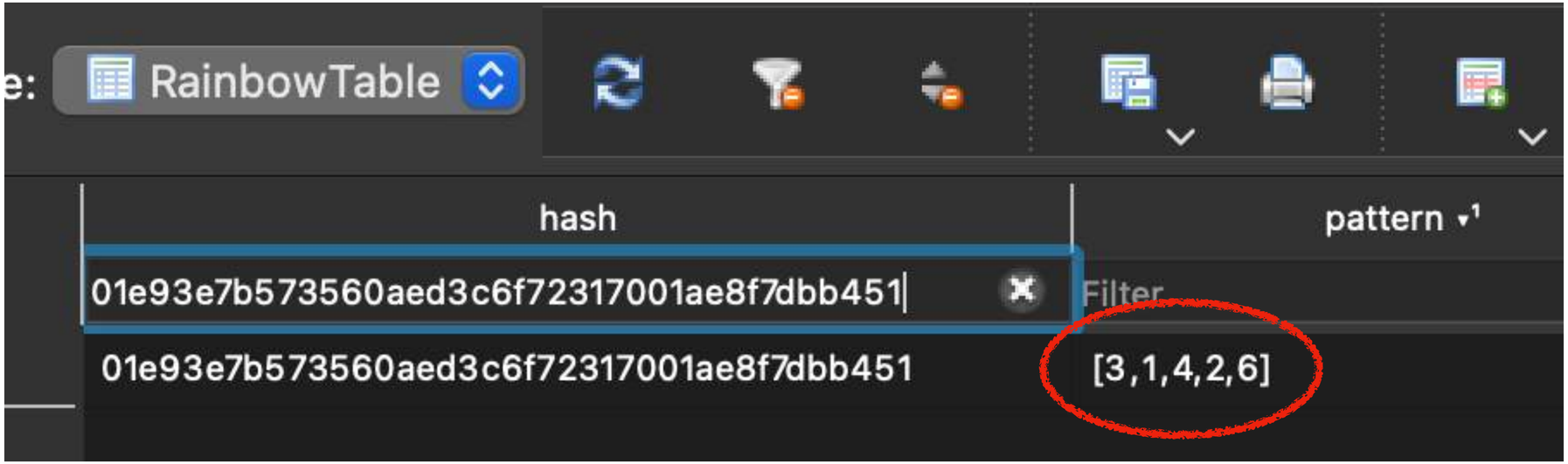
Crack pattern and pin

Pattern



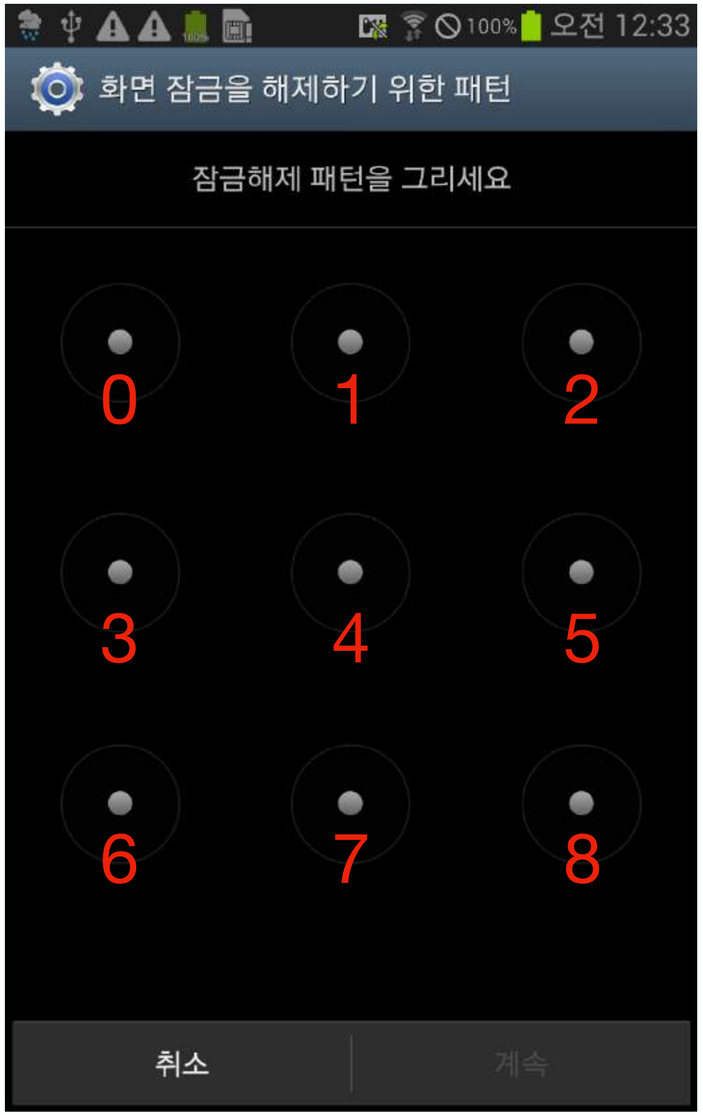
Crack pattern and pin

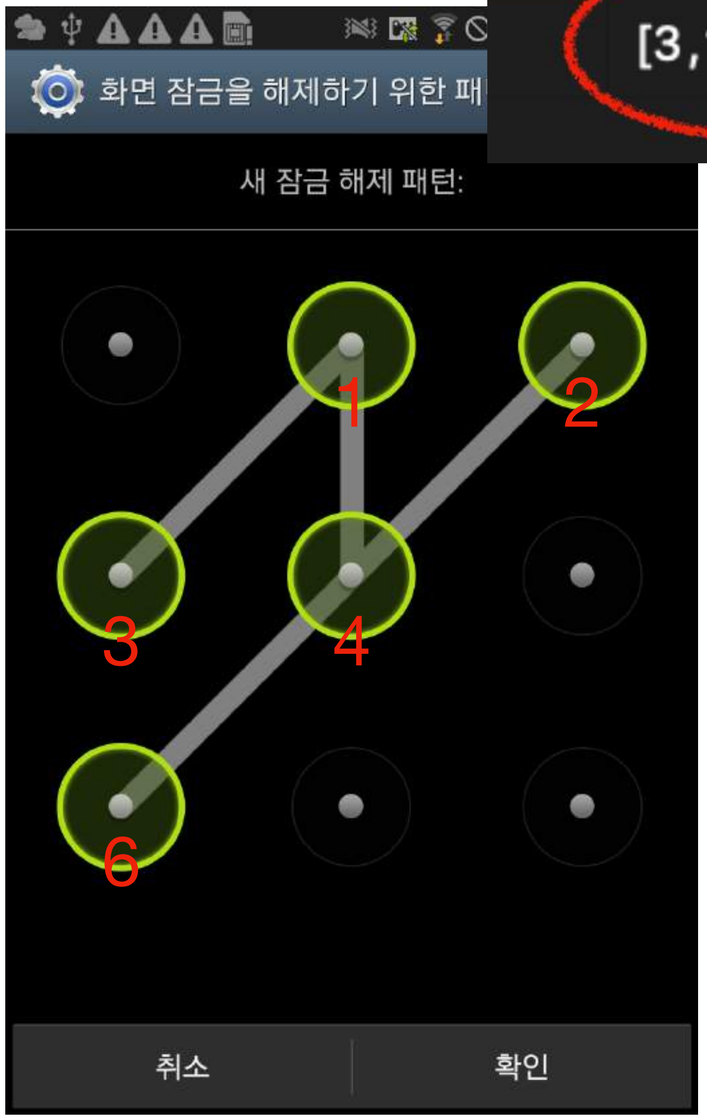
Pattern

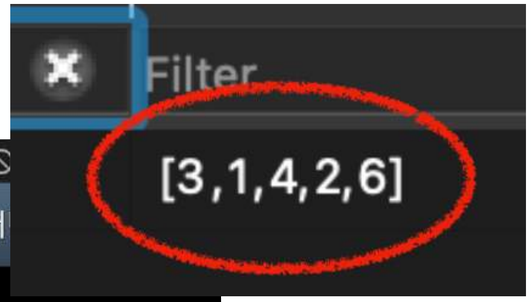


Crack pattern and pin

Pattern







Crack pattern and pin

Pin

```
 public byte[] passwordToHash(String password) {
        if (password == null) {
            return null;
        }
        String algo = null;
        byte[] hashed = null;
        try {
            byte[] saltedPassword = (password + getSalt()).getBytes();
            byte[] sha1 = MessageDigest.getInstance(algo = "SHA-1").digest(saltedPassword);
            byte[] md5 = MessageDigest.getInstance(algo = "MD5").digest(saltedPassword);
            hashed = (toHex(sha1) + toHex(md5)).getBytes();
        } catch (NoSuchAlgorithmException e) {
            Log.w(TAG, "Failed to encode string because of missing algorithm: " + algo);
        }
        return hashed;
}
```

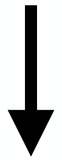
Crack pattern and pin

Pin

Hash salt is saved in `/data/system/locksettings.db`
-> convert to hexadecimal

`1136656D5C6718C1DEFC71B431B2CB5652A8AD550E20BDCF52B00002C8DF35C963B71298`

SHA-1 MD5



Crack md5 hash value using hashcat
(md5 cracking is usually faster than sha1)

Why old Android?

- This method only works under version 4.3.
- Let’s experiment with emulator!



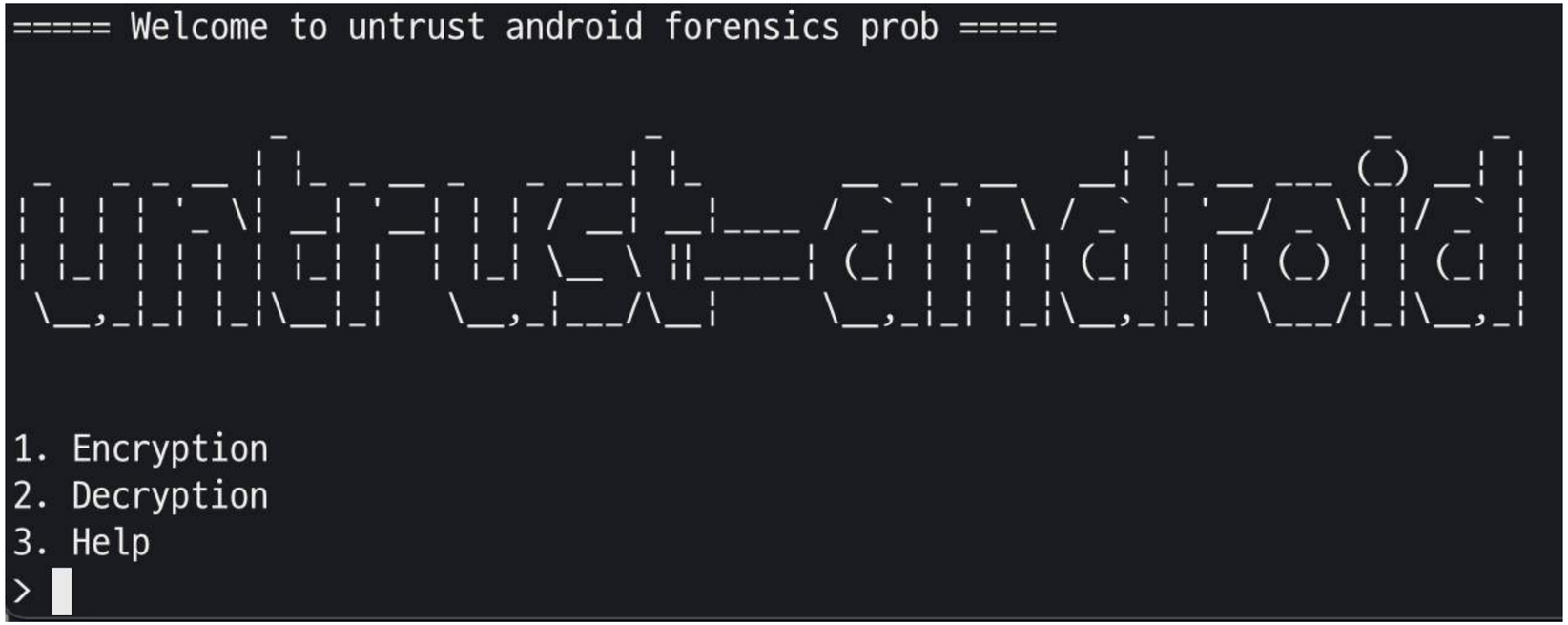
Why old Android?




```
~/Downloads/platform-tools                                  took 4s   at 14:57:1
❯ ./adb devices
List of devices attached
emulator-5554   device

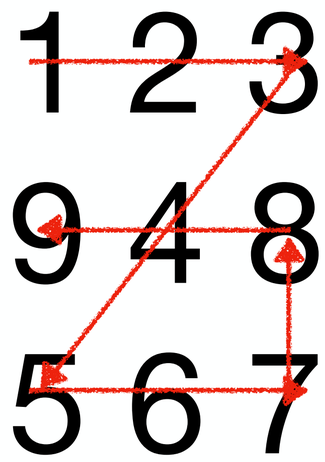

~/Downloads/platform-tools                                              at 14:57:1
❯ ./adb shell
emulator_arm64:/ $ su
emulator_arm64:/ # cd /data/syste
system/     system_ce/  system_de/
emulator_arm64:/ # cd /data/system/
emulator_arm64:/data/system # ls ¦ grep password.key
1¦emulator_arm64:/data/system # ls ¦ grep *.key
1¦emulator_arm64:/data/system # ls
appops                          ifw                         notification_policy.xml  slice
appops.xml                      install_sessions            overlays.xml             sound_model.db
battery-history                 install_sessions.xml        package-dcl.list         sound_model.db-journal
battery-saver                   integrity_rules             package-dex-usage.list   stats_pull
batterystats-daily.xml          integrity_staging           package-watchdog.xml     sync
batterystats.bin                job                         package_cache            syncmanager-log
blobstore                       last-fstrim                 packages-warnings.xml    uiderrors.txt
cachequota.xml                  last-header.txt             packages.list            unsolzygotesocket
device_policies.xml             locksettings.db             packages.xml             users
display-manager-state.xml       log-files.xml               procexitstore            watchlist_report.db
display_settings_freeform.xml   ndebugsocket               procstats                watchlist_report.db-journal
dropbox                         netpolicy.xml               recoverablekeystore.db   watchlist_settings.xml
entropy.dat                     netstats                    screen_on_time
graphicsstats                   notification_log.db         sensor_service
heapdump                        notification_log.db-journal shortcut_service.xml
emulator_arm64:/data/system #
```

Problem Solving



Problem Solving

Encryption





“[0, 1, 2, 4, 6, 7, 8, 5, 3]”



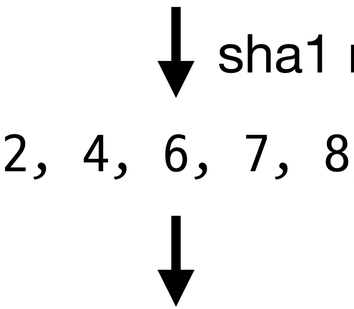
sha1

381aaf3c2d7704437b15d9f3fa8cf6a90b265c34

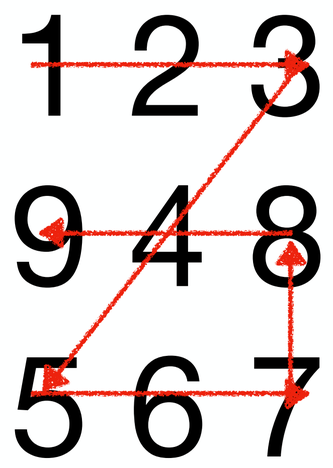
Problem Solving

Decryption

381aaf3c2d7704437b15d9f3fa8cf6a90b265c34



“[0, 1, 2, 4, 6, 7, 8, 5, 3]”



Problem Solving

pwntools

- `pip install pwntools`
- `p = remote(host, port)`
- `p.recvuntil(string)` : receives data until given string
- `p.sendline(string)` : send string + “\n”

`p.sendlineafter(str, str)`

- `p.recvline()` : receives one line
- `p.recv()` : receives until input
- `p.interactive()` : interactive shell

Problem Solving

```
nc untrust.kro.kr 31337
```

Problem Solving

Solution

```python
# !/usr/bin/python3
from pwn import *
import hashlib, sqlite3, json
p=remote("untrust.kro.kr", 31337)
p.sendlineafter("> ", "1")
```

Problem Solving

Solution

```
6   for prob in range(100): ## encryption prob
7       tmp=""
8       path=[]
9       for i in range(3):
10          tmp+=p.recvline().decode()
11      for i in tmp:
12          if i != "\n":
13              path.append(int(i))
14      queue=[]
15      for i in range(9):
16          queue.append(path.index(i+1))
17      p.sendlineafter("> ", hashlib.sha1(str(queue).encode()).hexdigest())
```

Problem Solving

Solution

```
p.sendlineafter("> ", "2")
conn = sqlite3.connect("/root/untrust-android/GestureRainbowTable.db")
cur = conn.cursor()
for prob in range(100): ## decryption prob
    hash = p.recvline().decode()[:-1]
    cur.execute(f"SELECT pattern FROM RainbowTable WHERE hash = '{hash}'")
    rows = cur.fetchall()
    for row in rows:
        queue=json.loads(row[0])
    path=[0]*9
    for i in range(9):
        path[queue[i]]=i+1
    correct_answer = ""
    for i in range(9):
        if (i+1)%3:
            correct_answer += str(path[i])
        else:
            correct_answer += str(path[i]) + "\n"
    correct_answer = correct_answer[:-1]
    for i in range(3):
        p.sendlineafter("> ", correct_answer.split("\n")[i])
p.interactive()
```

Problem Solving

Solution

```
⚡ root ~/untrust-android python3 solve.py
[+] Opening connection to untrust.kro.kr on port 31337: Done
[*] Switching to interactive mode
FLAG{4ndr0id+pwnt00ls=b3st_pr0b_3v3r!!}
[*] Got EOF while reading in interactive
$
```

FLAG{4ndr0id+pwnt00ls=b3st_pr0b_3v3r!!}

Thank you.